

Course Code: CS204 Advanced Algorithms and Data Structures
Assignment # 3: Graph Traversal and Dynamic Programming
Department of Computer Science
Semester: Spring 2024

Instructions:

Answer all 5 questions. Late submissions will receive a 20% penalty per day.
Submit your code implementations along with mathematical complexity proofs.

Question 1:

Implement Dijkstra shortest path algorithm using a Fibonacci heap.
Expected output should demonstrate $O(V \log V + E)$ asymptotic complexity.

Question 2:

Provide dynamic programming formulation for 0/1 Knapsack problem with runtime analysis.